

When using AI assisted tools to generate initial prototypes, "Requirements lost in AI conversations" and "Prototype → engineering handoff gap" are most commonly discussing issues. Is this true ? Need evidences

Yes there is growing evidence that two of the most commonly reported problems in AI-assisted prototyping workflows are:

1. **Requirements getting lost, distorted, or fragmented across long AI conversations**
2. **A gap between AI-generated prototypes and production-ready engineering handoff**

1. Evidence for “Requirements lost in AI conversations”

Evidence A — Replit PM Guide

[Replit: A Product Manager’s Guide to Using AI to Build Working Prototypes](#)

Replit explicitly states:

“Every handoff to design, engineering, and QA introduces translation loss...”

This is strong evidence that requirements degrade through AI-assisted workflows and conversational prototyping.

The article also argues that behavioral briefs and structured specs reduce this loss.

Evidence B — Requirements Engineering Research

Academic literature also supports this concern.

Systematic Mapping Study

[Requirements Engineering for Artificial Intelligence Systems \(arXiv\)](#)

Researchers found:

- existing requirements engineering methods are insufficient for AI systems
- current tooling struggles with explainability, ambiguity, and evolving requirements

Industry Practices Study

[Requirements Practices and Gaps When Engineering Human-Centered AI Systems \(arXiv\)](#)

The survey found:

- many teams still rely on informal tools like Office docs/UML
- current tools are “not equipped to manage AI-based software”

- requirements quality suffers in AI workflows

2. Evidence for “Prototype → engineering handoff gap”

This issue is even more widely discussed.

The pattern:

- AI tools generate impressive prototypes quickly
- engineering teams then struggle to productionize them
- prototypes lack architecture, data models, scalability, edge cases, security, accessibility, testing, etc.

Evidence A — “Prototype-to-Production Gap”

[Why AI-Generated UI Fails in Production](#)

This article explicitly calls it:

“the prototype-to-production gap”

It explains that AI tools optimize for:

- visual polish
- speed
- demoability

but not:

- maintainable architecture
- engineering constraints
- production readiness

Instead of using a systematic approach, AI generates hardcoded values. Design tokens are standardized, named values for foundational design properties like colors, spacing, and typography. You'll see `#333333` instead of `color-text-primary`. Why does this matter? Future updates become painful. Theme changes require hunting through every file. Visual consistency at scale becomes nearly impossible to maintain. What seemed fast at first slows everything down later.

